

## KUALA TRENGGANU, Malaysia

WMO: 486180

Lat: 5.3827N

Lon: 103.1080E

Elev: 6

StdP: 101.25

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 21.9       | 22.8 | 20.0                          | 14.7 | 24.6 | 20.8 | 15.5 | 24.9 | 7.4                   | 26.1 | 6.7 | 26.6 | 1.5                   | 200  | 0.328 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 7.3                    | 33.2            | 27.7 | 33.1 | 27.7 | 32.9 | 27.5 | 28.6                | 32.2 | 28.2 | 31.9 | 27.9 | 31.6 | 3.0                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.2                            | 23.1 | 31.3 | 27.1 | 22.9 | 31.2 | 27.0 | 22.7 | 31.1 | 92.4          | 32.6 | 90.7 | 32.3 | 89.3 | 31.6 | 31.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| 5.8               | 5.0  | 4.4 | DB                         | 21.2 | 33.8               | 0.9 | 0.7                                                | 20.5 | 34.3       | 20.0 | 34.7       | 19.4 | 35.1       | 18.8 | 35.7 | (4) |
|                   |      |     | WB                         | 20.3 | 29.5               | 1.0 | 0.7                                                | 19.6 | 30.0       | 19.0 | 30.4       | 18.4 | 30.8       | 17.7 | 31.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.7   | 26.7  | 27.1  | 27.7  | 28.5  | 28.8  | 28.5  | 28.2  | 28.1  | 27.9  | 27.5  | 27.0  | 26.8 |
|                                                                                  | DBStd       | 1.19   | 1.12  | 1.01  | 1.01  | 1.02  | 1.02  | 0.98  | 0.95  | 0.84  | 0.76  | 0.95  | 1.08  | 1.07 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD10.0     | 6474   | 517   | 479   | 548   | 554   | 583   | 555   | 563   | 560   | 537   | 543   | 512   | 522  |
|                                                                                  | CDD18.3     | 3432   | 259   | 246   | 289   | 304   | 324   | 305   | 305   | 302   | 287   | 285   | 262   | 264  |
|                                                                                  | CDH23.3     | 35859  | 2448  | 2518  | 3144  | 3564  | 3774  | 3378  | 3256  | 3179  | 2921  | 2786  | 2405  | 2486 |
|                                                                                  | CDH26.7     | 12792  | 555   | 733   | 1137  | 1493  | 1615  | 1390  | 1276  | 1237  | 1110  | 955   | 699   | 593  |
| Wind                                                                             | WSAvg       | 2.0    | 2.6   | 2.4   | 2.0   | 1.8   | 1.8   | 1.7   | 1.7   | 1.7   | 1.7   | 1.7   | 1.8   | 2.6  |
| Precipitation                                                                    | PrecAvg     | 2559   | 155   | 78    | 97    | 99    | 104   | 104   | 107   | 134   | 158   | 273   | 675   | 574  |
|                                                                                  | PrecMax     | 4592   | 1001  | 357   | 433   | 408   | 354   | 228   | 259   | 290   | 338   | 862   | 1596  | 1377 |
|                                                                                  | PrecMin     | 1724   | 12    | 1     | 0     | 0     | 6     | 22    | 23    | 32    | 24    | 92    | 225   | 72   |
|                                                                                  | PrecStd     | 621    | 226   | 82    | 96    | 86    | 82    | 54    | 55    | 66    | 80    | 162   | 383   | 367  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 31.2  | 31.3  | 32.9  | 34.0  | 34.2  | 34.0  | 33.2  | 33.1  | 33.1  | 32.6  | 32.0  | 31.2 |
|                                                                                  |             | MCWB   | 26.7  | 26.3  | 26.8  | 28.2  | 28.2  | 28.0  | 27.8  | 27.2  | 27.4  | 27.2  | 27.1  | 26.7 |
|                                                                                  | 2%          | DB     | 30.2  | 31.1  | 32.1  | 33.1  | 33.8  | 33.1  | 32.9  | 32.2  | 32.2  | 32.1  | 31.2  | 30.9 |
|                                                                                  |             | MCWB   | 26.2  | 26.2  | 26.5  | 27.8  | 28.0  | 27.7  | 27.5  | 27.2  | 27.0  | 26.9  | 26.9  | 26.6 |
|                                                                                  | 5%          | DB     | 30.0  | 30.2  | 31.7  | 32.8  | 33.1  | 32.9  | 32.1  | 32.1  | 32.0  | 31.2  | 30.9  | 30.1 |
|                                                                                  |             | MCWB   | 26.0  | 25.7  | 26.5  | 27.6  | 27.8  | 27.6  | 27.2  | 27.2  | 26.9  | 26.8  | 26.8  | 26.0 |
|                                                                                  | 10%         | DB     | 29.1  | 30.0  | 31.0  | 32.0  | 32.8  | 32.0  | 31.8  | 31.8  | 31.2  | 30.9  | 30.0  | 29.2 |
|                                                                                  |             | MCWB   | 25.4  | 25.6  | 26.4  | 27.1  | 27.6  | 27.2  | 27.1  | 27.1  | 26.8  | 26.7  | 26.3  | 25.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 27.6  | 27.0  | 28.0  | 29.0  | 28.7  | 28.8  | 28.7  | 29.0  | 28.2  | 28.2  | 27.7  | 27.2 |
|                                                                                  |             | MCDB   | 29.6  | 30.5  | 31.3  | 32.6  | 33.1  | 32.4  | 32.3  | 31.9  | 31.6  | 31.1  | 30.4  | 29.8 |
|                                                                                  | 2%          | WB     | 26.8  | 26.6  | 27.2  | 28.2  | 28.4  | 28.4  | 28.1  | 28.1  | 27.7  | 27.5  | 27.1  | 26.8 |
|                                                                                  |             | MCDB   | 29.4  | 30.0  | 30.8  | 32.3  | 32.7  | 32.0  | 31.7  | 31.4  | 31.3  | 30.8  | 30.1  | 29.4 |
|                                                                                  | 5%          | WB     | 26.4  | 26.3  | 26.8  | 27.7  | 28.0  | 28.0  | 27.7  | 27.6  | 27.3  | 27.1  | 26.8  | 26.5 |
|                                                                                  |             | MCDB   | 28.9  | 29.5  | 30.4  | 31.8  | 32.2  | 31.6  | 31.2  | 31.1  | 31.0  | 30.6  | 29.7  | 29.0 |
|                                                                                  | 10%         | WB     | 25.9  | 25.9  | 26.5  | 27.4  | 27.6  | 27.6  | 27.2  | 27.1  | 26.8  | 26.7  | 26.5  | 26.1 |
|                                                                                  |             | MCDB   | 28.3  | 29.0  | 29.9  | 31.3  | 31.6  | 31.1  | 30.7  | 30.7  | 30.4  | 29.9  | 29.3  | 28.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.9   | 5.8   | 6.5   | 7.0   | 7.3   | 7.4   | 7.3   | 7.4   | 7.3   | 6.6   | 5.6   | 4.8  |
|                                                                                  |             | MCDBR  | 5.8   | 6.2   | 7.3   | 7.6   | 7.9   | 7.8   | 7.8   | 7.7   | 7.7   | 7.3   | 6.7   | 5.8  |
|                                                                                  |             | MCWBR  | 2.7   | 2.7   | 3.0   | 3.3   | 3.6   | 3.7   | 4.0   | 4.0   | 3.8   | 3.4   | 3.1   | 2.8  |
|                                                                                  | 5% WB       | MCDBR  | 5.1   | 5.7   | 6.6   | 7.3   | 7.6   | 7.4   | 7.4   | 7.4   | 7.4   | 7.0   | 6.1   | 5.3  |
|                                                                                  | MCWBR       | 2.9    | 2.7   | 3.1   | 3.4   | 3.7   | 3.9   | 4.0   | 4.2   | 3.9   | 3.5   | 3.1   | 2.8   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.465  | 0.452 | 0.458 | 0.439 | 0.458 | 0.480 | 0.489 | 0.494 | 0.495 | 0.485 | 0.447 | 0.475 |      |
|                                                                                  | taud        | 2.306  | 2.328 | 2.329 | 2.383 | 2.340 | 2.287 | 2.256 | 2.226 | 2.224 | 2.260 | 2.383 | 2.289 |      |
|                                                                                  | Ebn at Noon | 857    | 877   | 871   | 872   | 836   | 806   | 802   | 813   | 826   | 837   | 867   | 840   |      |
|                                                                                  | Edh at Noon | 133    | 133   | 134   | 125   | 126   | 131   | 136   | 144   | 147   | 141   | 123   | 134   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.34   | 5.40  | 5.79  | 5.78  | 5.28  | 5.08  | 5.00  | 5.21  | 5.23  | 4.59  | 3.98  | 3.77  |      |
|                                                                                  | RadStd      | 0.38   | 0.42  | 0.48  | 0.30  | 0.28  | 0.23  | 0.29  | 0.14  | 0.27  | 0.28  | 0.29  | 0.38  |      |

## Historical Trends

|      | DBAvg                  | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (3 neighbors) | +0.20   | N/A    | N/A     | +0.30 | +0.52 | +0.45       | N/A     | N/A     | +65     |

Nomenclature: See separate page